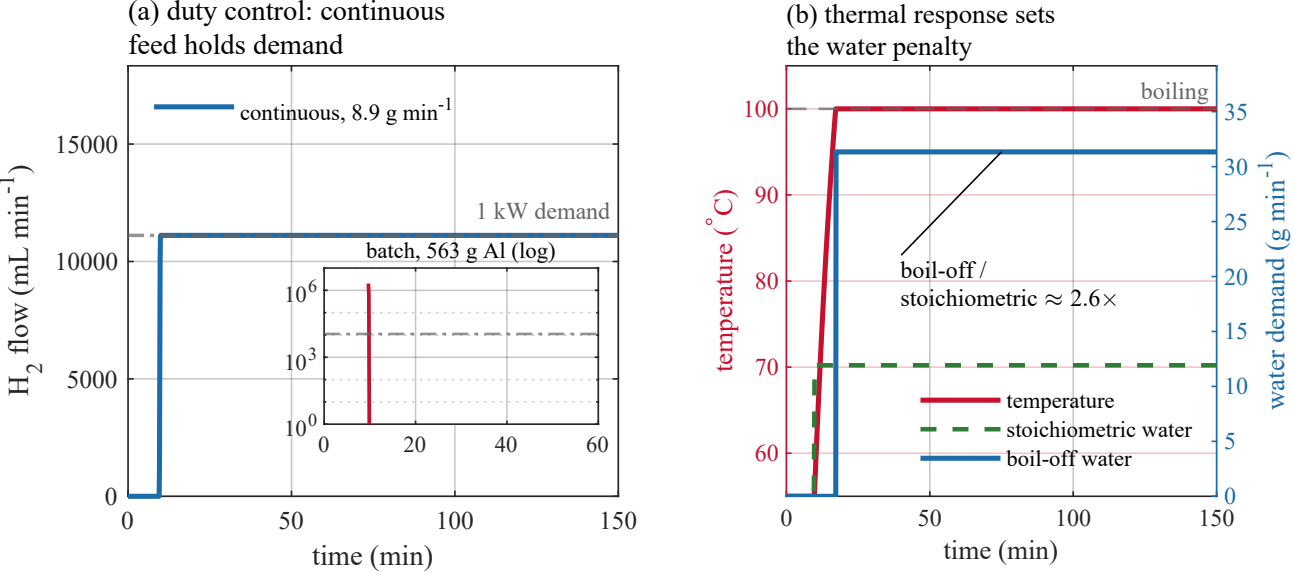

(a) duty control: continuous feed holds demand
continuous, 8.9 g min-1
1 kW demand
batch, 563 g Al (log)
H2 flow (mL min-1)
time (min)
(b) thermal response sets the water penalty
boiling
boil-off / stoichiometric ≈ 2.6×
temperature
stoichiometric water
boil-off water
temperature (°C)
water demand (g min-1)
time (min)